PULSE CODE MODULATION / 3.1

PCM: send a number for each sample

Measure, round and encode.

Pulse-code modulation (PCM). A method in which an analog signal is sampled, each sample is rounded to an allowed amplitude level, and the level is represented by a binary codeword.

The three main operations

1. **Sampling:** choose when to measure the signal.
2. **Quantization:** choose an allowed value close to the measured value.
3. **Encoding:** represent that value using bits, which are binary digits 0 and 1.

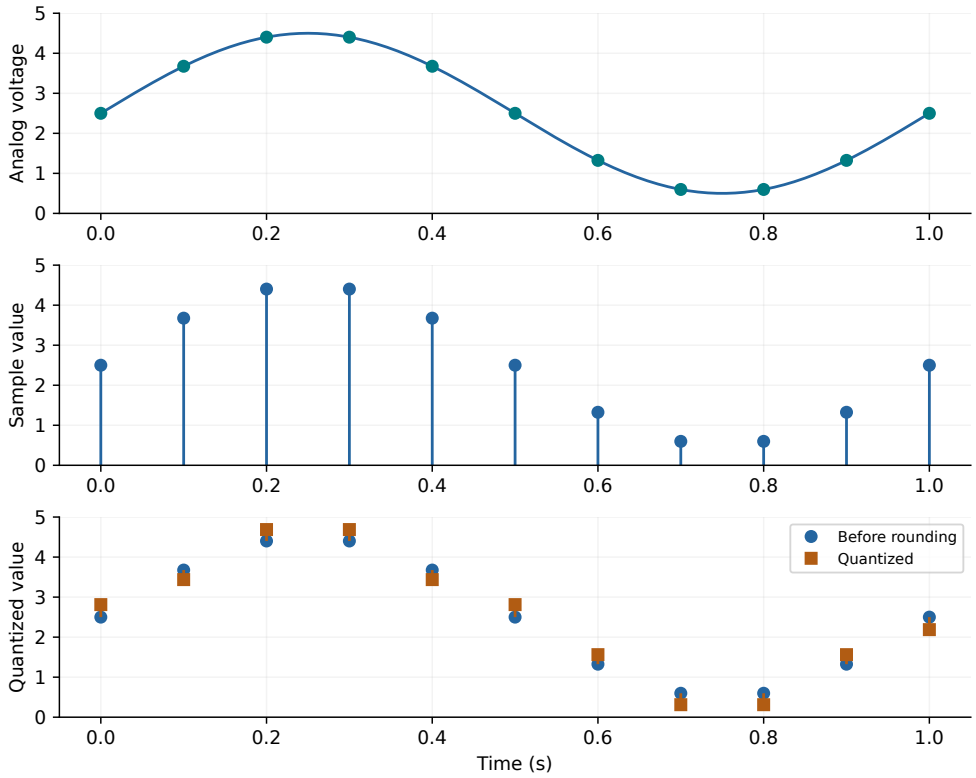


Figure 3.1: The analog signal, its samples and the rounded sample values.

Read the drawing. Sampling changes the set of measurement times. Quantization changes the measured amplitudes. The orange squares need not lie exactly on the blue sample values.

Explain why: Why is quantization needed before binary encoding?

A finite codeword can represent only a finite number of values. Quantization chooses one of those values for each sample.

PULSE CODE MODULATION / 3.2

PCM transmitter and receiver

Know the function of every block.

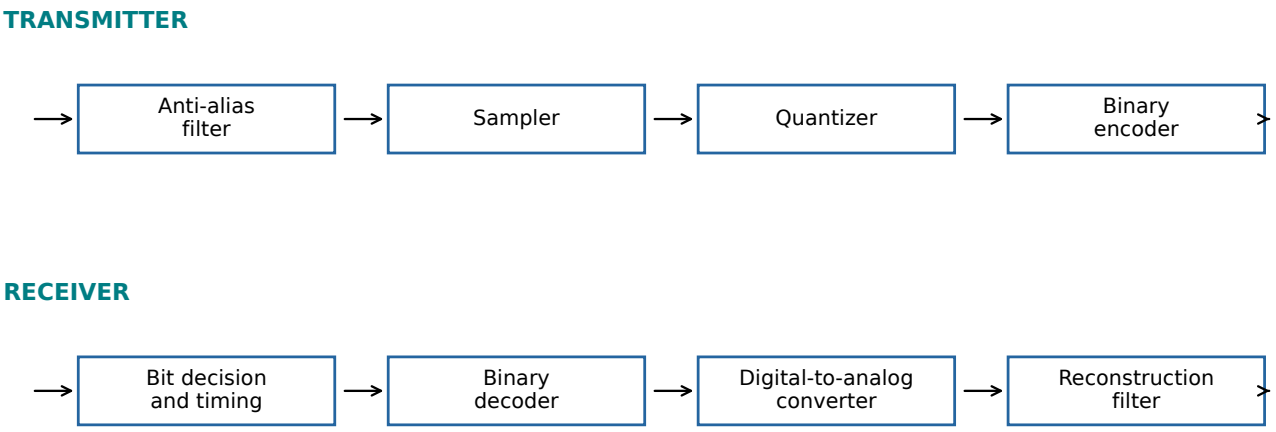


Figure 3.2: A complete PCM signal path. Sampling, bit decisions and decoding require timing information.

Transmitter functions

1. **Anti-alias filter:** removes unwanted high-frequency input components.
2. **Sampler:** measures the filtered signal at regular instants.
3. **Quantizer:** rounds each sample to an allowed level.
4. **Encoder:** assigns the binary word for that level.

Receiver functions

1. **Bit decision and timing:** identifies the received bits at the correct instants.
2. **Decoder:** converts each word back to its quantized sample value.
3. **Digital-to-analog converter (DAC):** produces an analog waveform from those values.
4. **Reconstruction filter:** smooths the waveform to recover the message band.

Explain why: Why does correct decoding not remove quantization error?

The codeword contains the rounded value. The original value within its quantization interval was not transmitted.

Draw and explain: Draw the two paths and state the function of each block.

Dr. Yassir Al-

PULSE CODE MODULATION / 3.3

Uniform quantization

Divide the input range into equal intervals.

Uniform quantizer. A quantizer whose decision intervals have the same width throughout its operating range. That width is the step size Δ.

Let $m_{\min}$ and $m_{\max}$ be the lower and upper range limits. Let L be the number of levels and v the number of bits per sample.

$$\boxed{L = 2^v}, \qquad \boxed{\Delta = \frac{m_{\max} - m_{\min}}{L}}.$$

More bits give more levels. More levels make the intervals smaller.

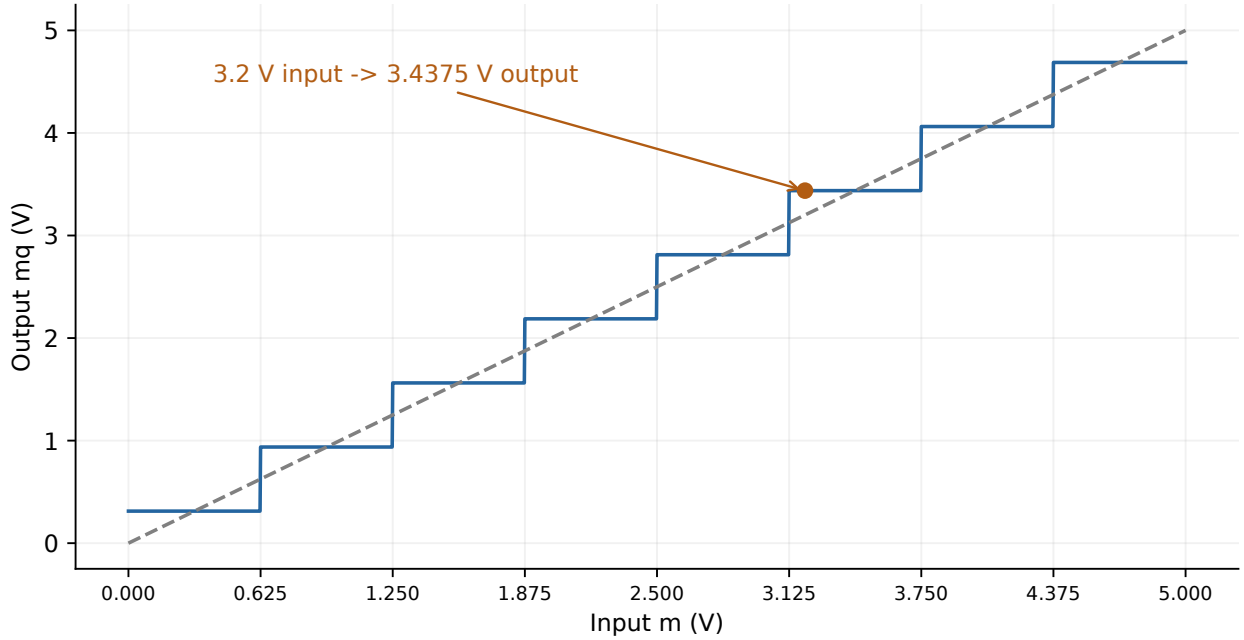


Figure 3.3: A 3-bit midrise quantizer covering 0 to 5 V. Each flat segment is one output value.

Read the figure. Move horizontally to the input value, then vertically to the staircase. The staircase height is the quantized output m_q.

Explain why: Why does increasing the number of levels improve resolution?

The same input range is divided into smaller intervals. Each rounded value can be closer to its input sample.

PULSE CODE MODULATION / 3.4

Midrise and midtread quantizers

Look at what happens near zero.

1. **Midrise:** zero is a decision boundary. The nearest output levels are $-\Delta/2$ and $+\Delta/2$.
2. **Midtread:** zero is an output level. A small interval around zero maps to zero.

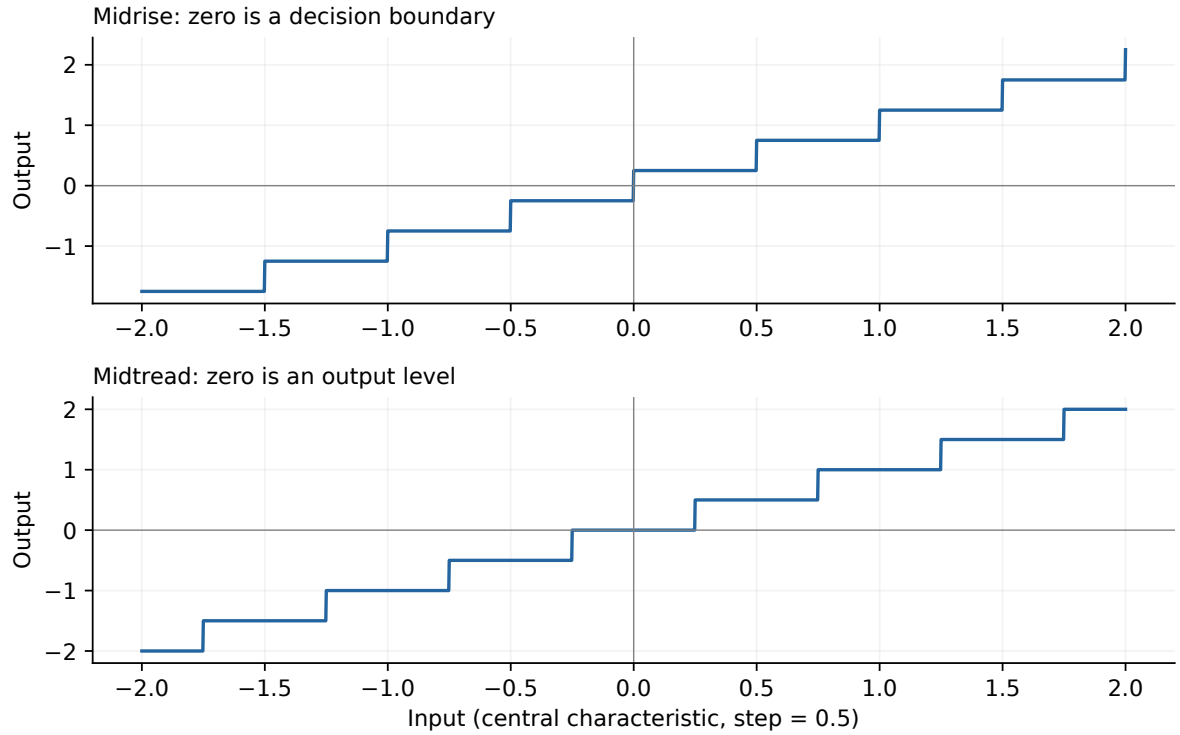


Figure 3.4: Central parts of the two quantizer characteristics. Finite-range end levels are not shown.

The numerical examples use a **midrise quantizer** unless stated otherwise. For an input inside the range, find its interval index i:

$$i = \left\lfloor \frac{m - m_{\min}}{\Delta} \right\rfloor, \qquad m_q = m_{\min} + \left(i + \frac{1}{2}\right)\Delta.$$

The floor symbol means round down to an integer. Indices run from 0 to $L - 1$. At the upper endpoint, use the final interval. Out-of-range inputs saturate and may have larger errors.

Try it. Does a symmetric midrise quantizer normally have an output level at zero?

Check: No. Zero separates its two central intervals.

PULSE CODE MODULATION / 3.5

A 3-bit ADC: solve it slowly

Use the input range 0 to 5 V and the sample 3.2 V.

Analog-to-digital converter (ADC). A device that converts analog input values into digital codes. Its resolution is commonly specified by the number of output bits.

Example 3.1 / Levels, step size, output and code

The converter has $\nu = 3$ bits. Find its number of levels:

$$L = 2^3 = 8.$$

Divide the 5 V range into eight equal intervals:

$$\Delta = (5 - 0)/8 = 0.625\,\text{V}.$$

Find the interval containing $m = 3.2$ V:

$$i = \lfloor 3.2/0.625 \rfloor = \lfloor 5.12 \rfloor = 5.$$

Interval 5 extends from $5(0.625) = 3.125$ V to $6(0.625) = 3.750$ V. Use its midpoint:

$$m_q = (5 + 0.5)(0.625) = 3.4375\,\text{V}.$$

Write the index 5 as a three-bit binary word:

$$\boxed{5 = 101_2}.$$

Result. The input 3.2 V is represented by the code 101 and reconstructed as 3.4375 V.

Explain why: Why is the output 3.4375 V rather than the lower boundary 3.125 V?

A midrise rounding quantizer uses the midpoint of the selected interval. A boundary is a decision threshold, not its reconstruction value.

Try it. For a 1.4 V input, find the index and output.

Check: $i = 2$, $m_q = 1.5625$ V and code 010.

PULSE CODE MODULATION / 3.6

The code table and transmitted bits

The code identifies a level, not an exact original voltage.

Input interval (V)	Output m_q (V)	Code
$[0, 0.625)$	0.3125	000
$[0.625, 1.250)$	0.9375	001
$[1.250, 1.875)$	1.5625	010
$[1.875, 2.500)$	2.1875	011
$[2.500, 3.125)$	2.8125	100
$[3.125, 3.750)$	3.4375	101
$[3.750, 4.375)$	4.0625	110
$[4.375, 5.000]$	4.6875	111

A square bracket includes the endpoint. A round bracket excludes it. The final upper endpoint is included by convention.

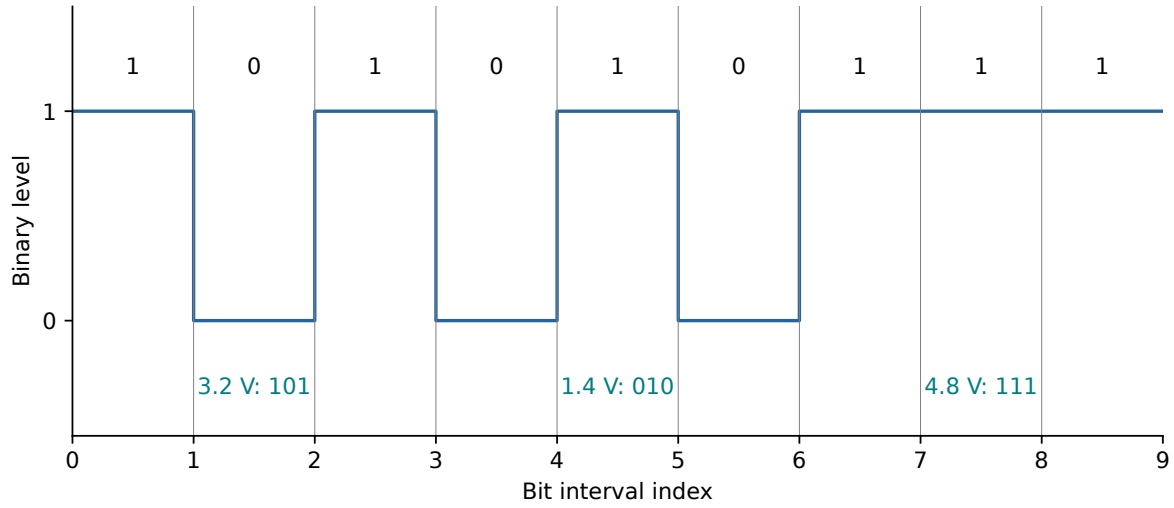


Figure 3.5: Three samples become three words: 101, 010 and 111. The drawing uses a simple binary pulse representation.

Read the drawing. Each sample uses three consecutive bit intervals. The next sample starts after those three bits. The binary code assignment here is straight binary, most significant bit first.

Explain why: Why do 3.2 V and 3.6 V produce the same code?

Both lie in interval 5. That entire interval maps to one reconstruction value and one codeword.

PULSE CODE MODULATION / 3.7

Quantization error

Subtract the rounded value from the original sample.

Quantization error. The difference between an input sample and its quantized value. In this lecture, $e = m - m_q$.

For rounding within the operating range,

$$\boxed{|e| \le \Delta/2}.$$

Example 3.2 / Error for the 3.2 V sample

Use $m_q = 3.4375$ V from the preceding example:

$$e = 3.2 - 3.4375 = -0.2375\,\text{V}.$$

The negative sign means the rounded value is higher than the input. Check its magnitude:

$$|e| = 0.2375\,\text{V} < 0.625/2 = 0.3125\,\text{V}.$$

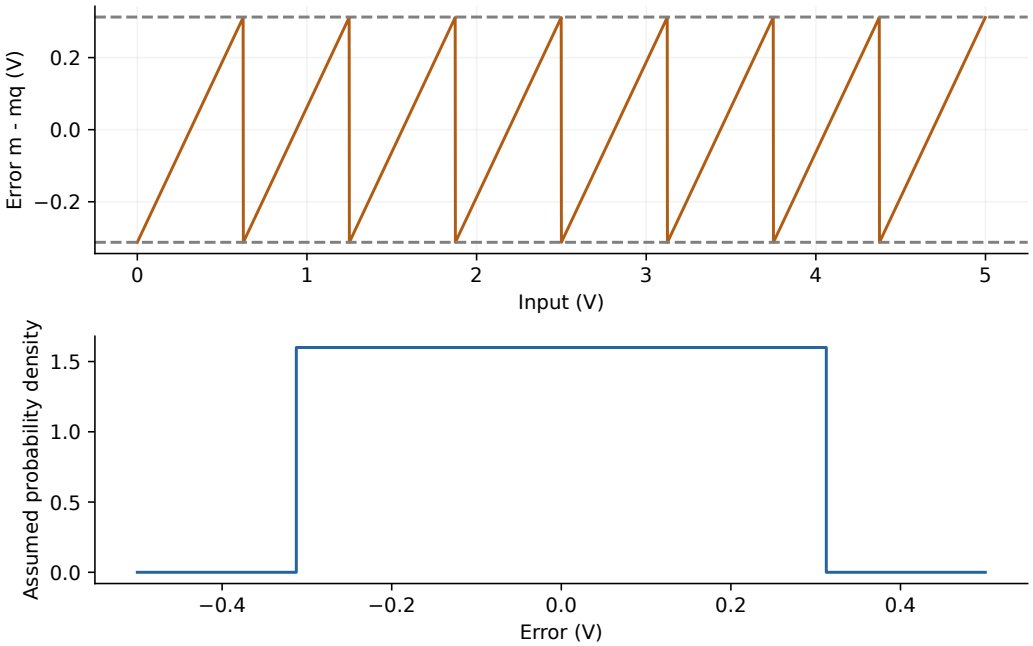


Figure 3.6: Upper panel: rounding error across the range. Lower panel: the uniform error-density model used for noise calculations.

Explain why: Why is the maximum rounding error half a step?

The reconstruction value is at the interval centre. The furthest point inside that interval is a boundary, half a step away.

PULSE CODE MODULATION / 3.8

Quantization noise power

Use a statistical model for many rounded samples.

When the signal spans many levels and the error behaves approximately uniformly between $-\Delta/2$ and $\Delta/2$, its mean-square value is

$$\boxed{N_q = E[e^2] \approx \frac{\Delta^2}{12}}.$$

$E[\,]$ means average or expected value. If samples are voltages, N_q is in V^2. To obtain watts across a resistance R, divide by R.

Simple derivation

The assumed error density is $1/\Delta$ over an interval of width Δ. Multiply squared error by this density and add all contributions:

$$\begin{aligned} N_q &= \int_{-\Delta/2}^{\Delta/2} e^2 \frac{1}{\Delta}\, de \\ &= \frac{1}{\Delta}\left[\frac{e^3}{3}\right]_{-\Delta/2}^{\Delta/2} = \frac{\Delta^2}{12}. \end{aligned}$$

Example 3.3 / Noise estimate for the 3-bit converter

Use $\Delta = 0.625$ V:

$$N_q \approx 0.625^2/12 = 0.032552\ \mathrm{V}^2.$$

The root-mean-square (RMS) error is $\sqrt{N_q} = 0.1804$ V. This is a statistical estimate over many samples, not the error of every individual sample.

Explain why: Why must the noise-model assumptions be stated?

A fixed input or a signal using very few levels can produce structured rounding errors. Their mean-square value need not equal $\Delta^2/12$.

PULSE CODE MODULATION / 3.9

Two sources of error in PCM

Quantization error and channel errors have different causes.

1. **Quantization error:** introduced when the transmitter rounds sample amplitudes. It remains even if every bit arrives correctly.
2. **Transmission error:** introduced when noise or distortion causes a receiver to decide the wrong bit.

> **Regeneration.** Detecting the bits, restoring their pulse shapes and restoring timing before retransmission.

Regeneration can prevent gradual noise accumulation when bit decisions are correct. It does not recover a bit already decided incorrectly.

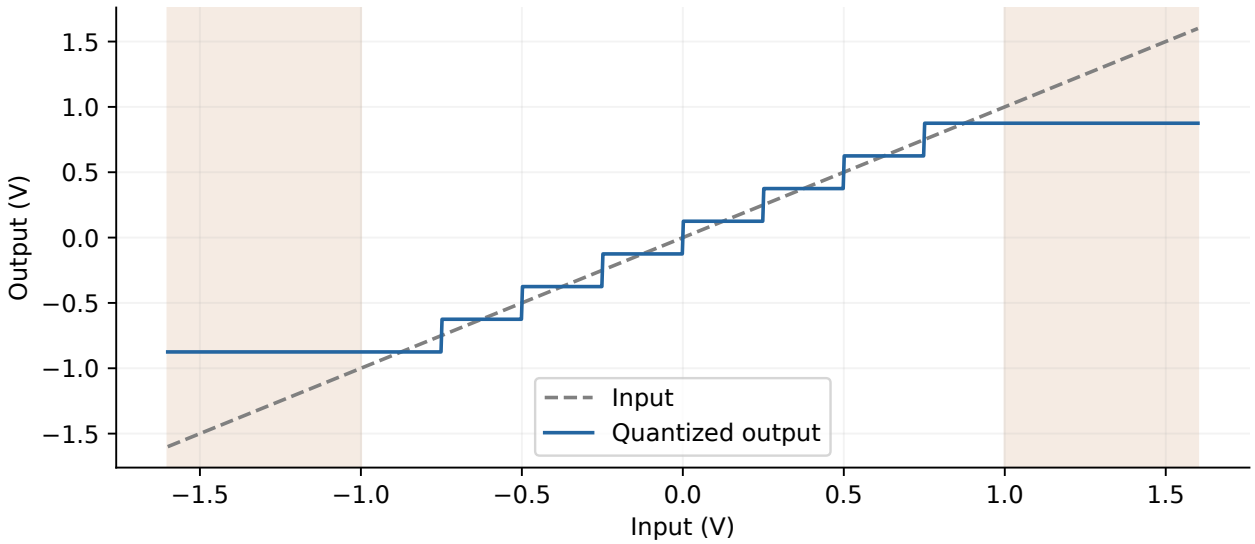


Figure 3.7: When the input exceeds the quantizer range, the output saturates. The shaded regions show overload.

> **Explain why: Why can the error exceed $\Delta/2$ during overload?**
>
> The input is outside the specified rounding range. The output remains at an end level while the input continues to increase.

Advantages and limitations of PCM

1. Bits can be regenerated and processed using digital circuits.
2. Different information sources can share a common digital representation.
3. Error-control coding and encryption can be added. PCM alone does not provide secrecy.
4. More bits improve amplitude resolution but increase bit rate and required channel resources.

PULSE CODE MODULATION / 3.10

Bits, sampling rate and bandwidth

Count how many bits are sent in one second.

Let f_s be samples per second and ν be bits per sample. With no overhead, the bit rate r is

$$\boxed{r = \nu f_s}, \qquad \boxed{T_b = 1/r},$$

where T_b is the duration of one bit. For L required levels, $\nu = \lceil \log_2 L \rceil$: round up because a word must have enough codes. For ideal binary baseband transmission with zero intersymbol interference and zero roll-off, the minimum one-sided channel bandwidth is

$$\boxed{B_{\min} = r/2}.$$

With raised-cosine roll-off α, $B = (1+\alpha)r/2$. Actual bandwidth depends on pulse shaping. Rectangular bit pulses do not have a strict finite bandwidth.

Example 3.4 / Eight-bit voice PCM

Use $f_s = 8000$ samples/s and $\nu = 8$:

$$r = 8(8000) = 64000\,\text{bit/s} = 64\,\text{kbit/s}.$$

One bit lasts $T_b = 1/64000 = 15.625\,\mu\text{s}$. The ideal minimum bandwidth is 32 kHz.

Explain why: Why does one extra bit per sample increase the bit rate?

That extra bit must be transmitted for every sample. With f_s fixed, it adds f_s bits each second.

Try it. Use $f_s = 6000$ and $\nu = 4$. Find r and ideal $B_{\min}$.

Check: 24 kbit/s and 12 kHz.

PULSE CODE MODULATION / 3.11

Signal-to-quantization-noise ratio

Compare the useful mean-square value with the rounding error.

SQNR. Signal-to-quantization-noise ratio is the signal mean-square value divided by the mean-square quantization error. It excludes other noise sources.

Let $P = E[m^2]$ in V^2. Use the same units in numerator and denominator:

$$\boxed{\mathrm{SQNR} = P/N_q}, \qquad \mathrm{SQNR}_{\mathrm{dB}} = 10\log_{10}(P/N_q).$$

For a symmetric range $-m_p$ to $+m_p$ and $L = 2^\nu$,

$$\Delta = \frac{2m_p}{L}, \qquad N_q \approx \frac{m_p^2}{3L^2}.$$

Substituting gives the useful general formula:

$$\boxed{\mathrm{SQNR} \approx \frac{3P}{m_p^2}\,2^{2\nu}}.$$

Do not automatically set $P = 1$. It depends on the signal amplitude and waveform.

Example 3.5 / A specified mean-square value

Use the range -5 to $+5$ V, $\nu = 8$ and $P = 2\ \mathrm{V}^2$.

$$L = 256, \quad \Delta = 10/256 = 0.0390625\,\mathrm{V}.$$

$$N_q \approx \Delta^2/12 = 0.000127157\,\mathrm{V}^2.$$

$$\mathrm{SQNR} \approx 2/0.000127157 = 15728.64.$$

$$\mathrm{SQNR}_{\mathrm{dB}} \approx 41.97\,\mathrm{dB}.$$

For a 3 kHz message sampled at the theoretical boundary, $f_s = 6$ kHz and $r = 48$ kbit/s. A practical design uses a sampling margin.

Dr. Yassir Al-Ameen

PULSE CODE MODULATION / 3.12

The 6 dB per bit rule

Use this formula for a full-scale sine wave.

For a sine wave with peak amplitude m_p, the mean-square value is $P = m_p^2/2$. With a matching quantizer range $[-m_p, m_p]$,

$$\mathrm{SQNR} \approx \frac{m_p^2/2}{m_p^2/(3L^2)} = \frac{3}{2}L^2.$$

Convert to decibels and use $L = 2^\nu$:

$$\boxed{\mathrm{SQNR}_{\mathrm{dB}} \approx 1.76 + 6.02\nu}.$$

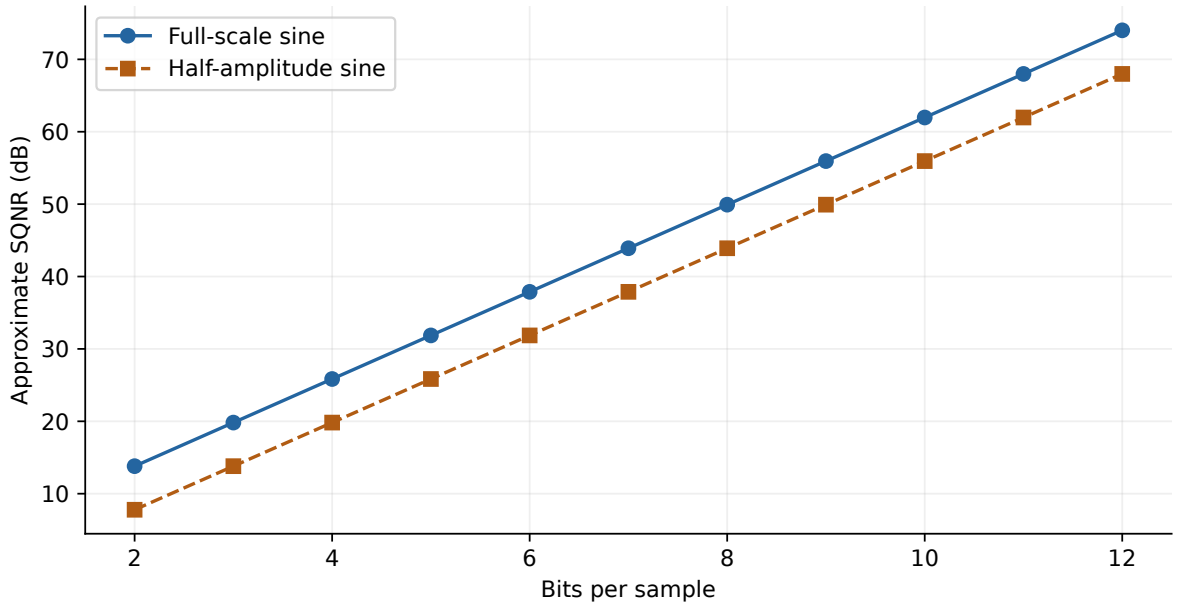


Figure 3.8: Each extra bit gives about 6 dB under the uniform-error model. A half-amplitude sine is about 6 dB worse with the same quantizer range.

Explain why: Why does adding one bit improve SQNR by about 6 dB?

One bit doubles the levels, halves the step and reduces $\Delta^2/12$ by a factor of four. The improvement is $10\log_{10} 4 = 6.02$ dB.

Example 3.6 / A 10-bit sinusoidal input

For $m(t) = 3\cos(500\pi t)$ V with a matching ± 3 V range,

$$\mathrm{SQNR}_{\mathrm{dB}} \approx 1.76 + 6.02(10) = 61.96\,\mathrm{dB}.$$

Its frequency is 250 Hz. The SQNR calculation assumes no overload and the usual quantization-noise model.

PULSE CODE MODULATION / 3.13

Choose enough bits for the required quality

Round up when the requirement is a minimum.

Example 3.7 / At least 40 dB for a full-scale sine

Start with the required inequality:

$$1.76 + 6.02\nu \geq 40.$$

Subtract 1.76 and divide by 6.02:

$$\nu \geq (40 - 1.76)/6.02 = 6.352.$$

Use the next whole number: $\nu = 7$. Then $L = 128$ and predicted SQNR is 43.90 dB. Six bits give 37.88 dB, which does not meet the requirement.

Example 3.8 / A signal that is not a full-scale sine

Use range ± 3.8 V, signal mean-square value $P = 0.03\ \text{V}^2$ and required SQNR 20 dB. This corresponds to 30 mW across a 1-ohm reference load. Convert the ratio to linear form: $10^{20/10} = 100$.

$$\frac{3(0.03)L^2}{3.8^2} \geq 100 \quad \Longrightarrow \quad L \geq \sqrt{\frac{100(3.8)^2}{0.09}} = 126.67.$$

Binary words require $\nu = \lceil \log_2(126.67) \rceil = 7$, giving 128 levels.

Explain why: Why is the full-scale sine formula unsuitable for every waveform?

Its constant 1.76 dB comes from the sine-wave mean-square value. A different waveform or a smaller amplitude changes the signal power.

PULSE CODE MODULATION / 3.14

Accuracy and limited bit rate

A minimum accuracy and a maximum rate require different rounding.

Example 3.9 / Accuracy of 0.1 percent of peak full scale

Use a ± 10 V range and $f_m = 100$ Hz. Define the error limit as 0.1% of the 10 V peak, so $|e| \leq 0.01$ V. Since $|e| \leq \Delta/2$, require $\Delta \leq 0.02$ V. Thus

$$L \geq 20/0.02 = 1000, \qquad \nu = 10, \quad L = 1024.$$

At the theoretical sampling boundary, $f_s = 200$ samples/s. Hence $r = 2000$ bit/s and $B_{\min} = 1000$ Hz. If full scale instead means the entire 20 V span, the allowed error is 0.02 V and nine bits suffice. Always define the percentage reference.

Example 3.10 / A 50 kbit/s channel

A 3.5 kHz message has peak-to-peak range 4 V and RMS value 0.2 V. At $f_s = 7$ kHz, the bit-rate constraint gives

$$\nu \leq 50000/7000 = 7.142.$$

Choose seven bits, not eight. Eight would require 56 kbit/s. With $m_p = 2$ V and $P = 0.2^2 = 0.04$ V^2,

$$\mathrm{SQNR} \approx \frac{3(0.04)128^2}{2^2} = 491.52, \quad \mathrm{SQNR}_{\mathrm{dB}} \approx 26.92\,\mathrm{dB}.$$

At seven bits, sampling may be increased up to $50/7 = 7.143$ kHz while staying within the rate limit, if no overhead is added.

Dr. Y

PULSE CODE MODULATION / 3.15

More rate and bandwidth problems

Keep units consistent before substituting.

Example 3.11 / A 7-bit system at 50 Mbit/s

Find the sampling rate: $f_s = r/\nu = 50 \times 10^6/7 = 7.143$ Msamples/s. The theoretical upper message bandwidth is $f_s/2 = 3.571$ MHz. A practical message band is smaller. A full-scale 1 MHz sine has predicted SQNR $1.76 + 6.02(7) = 43.90$ dB.

Example 3.12 / Television signal with 512 levels

Use message bandwidth $f_m = 10.2$ MHz and $L = 512$. Find the word length: $\nu = \log_2(512) = 9$ bits. At the theoretical boundary, $f_s = 20.4$ Msamples/s.

$$r = 9(20.4) = 183.6\,\text{Mbit/s}, \qquad B_{\min} = 91.8\,\text{MHz}.$$

For a full-scale sinusoidal test input, predicted SQNR is 55.94 dB. Bandwidth alone does not specify the SQNR of an arbitrary television waveform.

Example 3.13 / Voice over a 36 kbit/s channel

Use $f_m = 3.2$ kHz. The theoretical minimum sample rate is 6.4 kHz.

$$\nu \leq 36000/6400 = 5.625.$$

Choose five bits and 32 levels. If the entire bit rate is used, $f_s = 36000/5 = 7200$ samples/s, which exceeds 6400.

PULSE CODE MODULATION / 3.16

Check the specification before accepting a design

A numerical answer must satisfy every requirement.

Example 3.14 / Sixteen levels and a one-percent error limit

A 3 kHz message uses a symmetric range $\pm m_p$ and $L = 16$ levels. It also requires maximum rounding error no larger than 1% of m_p. For 16 levels, $\Delta = 2m_p/16$ and maximum error is $m_p/16 = 6.25\%$ of m_p. Therefore 16 levels do not meet the one-percent condition. To meet that condition, require $m_p/L \leq 0.01m_p$, so $L \geq 100$. Binary encoding then needs seven bits and 128 levels. At the theoretical sampling boundary $f_s = 6$ kHz, the corrected design needs $r = 42$ kbit/s. The 16-level design uses 24 kbit/s but fails the accuracy requirement.

Example 3.15 / Reduced sine-wave amplitude

A 7-bit PCM system has a ± 5 V quantizer range and rate 56 kbit/s. Its sample rate is $56000/7 = 8000$ samples/s, allowing a theoretical 4 kHz message limit. For a sine of peak amplitude a, use

$$\mathrm{SQNR}_{\mathrm{dB}} \approx 43.90 + 20\log_{10}(a/5).$$

To keep SQNR at least 30 dB,

$$a \geq 5\,10^{(30-43.90)/20} \approx 1.009\,\mathrm{V}.$$

Thus the peak amplitude may range approximately from 1.009 to 5 V under the model without overload.

Dr. Yassir

PULSE CODE MODULATION / 3.17

Non-uniform quantization

Use smaller steps where fine detail matters most.

Non-uniform quantization. Quantization with unequal decision-interval widths across the input range. Speech-oriented designs commonly use smaller intervals near zero and larger intervals at high amplitudes.

Why use it?

1. A small signal can be badly affected by a fixed absolute rounding error.
2. Smaller intervals near zero reduce the error of low-amplitude samples.
3. Larger intervals at high amplitudes trade absolute precision for a wider useful range.

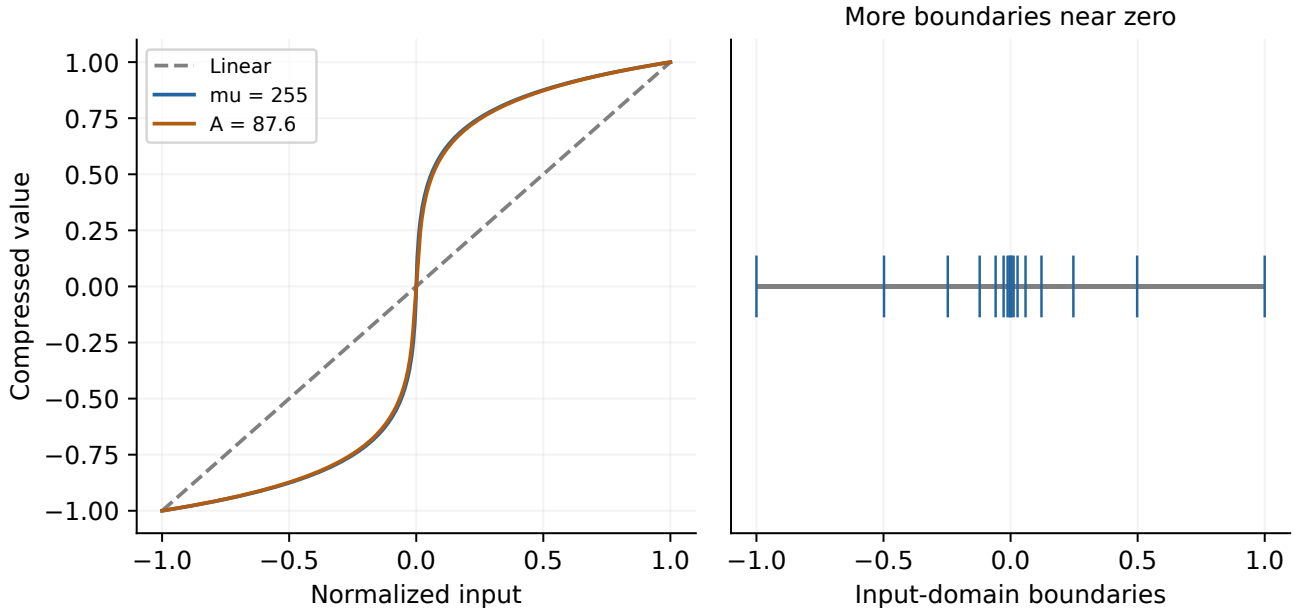


Figure 3.9: Compression curves and input-domain boundaries. More boundaries are allocated near zero.

Explain why: Why is uniform quantization less efficient for weak speech samples?

A fixed step can be large compared with the sample amplitude. The same absolute error then produces a larger relative error.

Non-uniform quantization does not remove noise. Its benefit depends on the input amplitude distribution and the selected compression law.

PULSE CODE MODULATION / 3.18

Companding: compress, then expand

Use a uniform quantizer in a transformed amplitude scale.

Companding. Compressing the amplitude range before quantization and applying the inverse expansion after decoding. The name combines compressing and expanding.

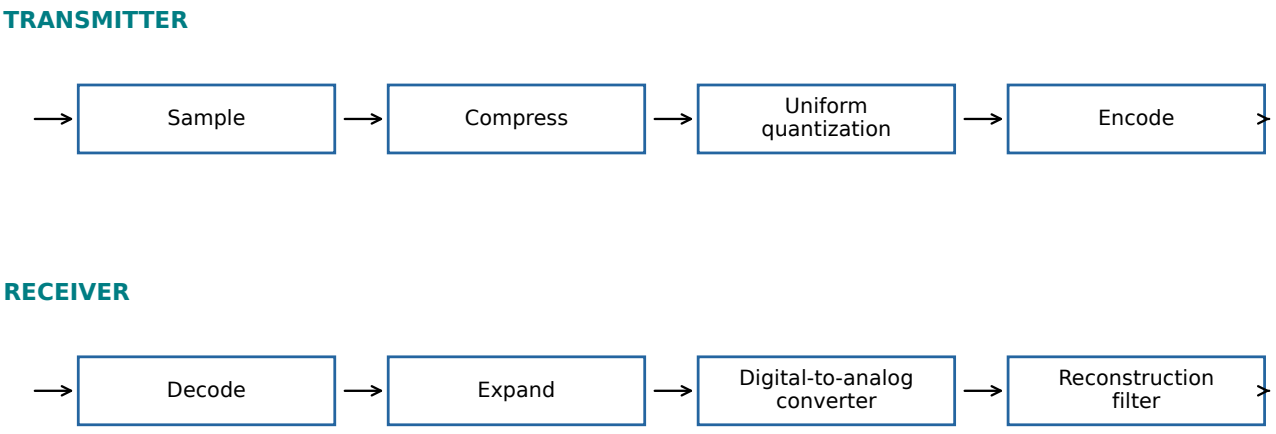


Figure 3.10: Conceptual companding PCM chain. The compressor and expander operate on sample values in this model.

Functions of the blocks

1. **Compressor:** maps small amplitudes to a larger fraction of the available scale.
2. **Uniform quantizer:** rounds values on that compressed scale.
3. **Encoder and decoder:** transmit and recover the quantized compressed values.
4. **Expander:** applies the inverse mapping to return to the original amplitude scale.

Explain why: Why does uniform quantization after compression act like non-uniform quantization at the input?

Equal steps on the compressed scale correspond to unequal intervals on the original scale. Near zero those input intervals are smaller.

Explain why: Why does expansion not exactly recover every input sample?

Compression can be inverted, but the intervening rounding has discarded information. Expansion cannot recover that lost information.

Draw and explain: Draw both paths and identify where rounding occurs.

PULSE CODE MODULATION / 3.19

The mu-law compressor

Normalize the sample before using the formula.

Let $x = m/m_p$ be a normalized input with $|x| \leq 1$. Let y be its compressed value. The continuous mu-law is

$$\boxed{y = \operatorname{sgn}(x)\frac{\ln(1+\mu|x|)}{\ln(1+\mu)}}.$$

$\operatorname{sgn}(x)$ keeps the sign, and ln is the natural logarithm. A common parameter is $\mu = 255$. The inverse expansion is

$$\boxed{x = \operatorname{sgn}(y)\frac{(1+\mu)^{|y|}-1}{\mu}}.$$

Example 3.16 / Compress a small sample

Use $x = 0.1$ and $\mu = 255$:

$$y = \frac{\ln(1+255(0.1))}{\ln 256} = \frac{\ln 26.5}{\ln 256} \approx 0.59099.$$

A sample that occupied 10% of the input scale now occupies about 59.1% of the compressed scale. This is a nonlinear mapping, not extra information. If no quantization intervenes, expanding the unrounded value returns $x = 0.1$. With quantization, the reconstructed value is only approximate.

The continuous formula explains the principle. Actual G.711 mu-law coding uses a specified segmented digital mapping. Its exact codewords should not be inferred by directly applying a generic straight-binary encoder to this formula.

Try it. What happens to a negative input of -0.1?

Check: Its compressed value is approximately -0.59099.

PULSE CODE MODULATION / 3.20

The A-law compressor

Use a linear part near zero and a logarithmic part outside it.

For normalized input $|x| \le 1$ and parameter $A > 1$, define

$$y = \operatorname{sgn}(x) \begin{cases} \dfrac{A|x|}{1+\ln A}, & 0 \le |x| < 1/A, \\ \dfrac{1+\ln(A|x|)}{1+\ln A}, & 1/A \le |x| \le 1. \end{cases}$$

A common parameter is $A = 87.6$. First compare $|x|$ with $1/A$, then select the correct expression. The inverse is

$$x = \operatorname{sgn}(y) \begin{cases} \dfrac{|y|(1+\ln A)}{A}, & |y| < \dfrac{1}{1+\ln A}, \\ \dfrac{\exp(|y|(1+\ln A)-1)}{A}, & \dfrac{1}{1+\ln A} \le |y| \le 1. \end{cases}$$

Example 3.17 / Select the correct branch

For $x = 0.01$, $1/A \approx 0.01142$. Since $0.01 < 0.01142$, use the linear branch:

$$y = \frac{87.6(0.01)}{1+\ln 87.6} \approx 0.1601.$$

For $x = 0.1$, use the logarithmic branch, giving $y \approx 0.5793$.

Explain why: Why is A-law written as two expressions?

It uses a linear region near zero and a logarithmic region at larger magnitudes. The expressions meet continuously at the boundary.

G.711 specifies eight bits per sample at 8000 samples/s, giving 64 kbit/s before framing overhead. Its exact A-law and mu-law code mappings are defined by the standard.

PULSE CODE MODULATION / 3.21

Uniform and companded PCM: compare carefully

A weak-signal benefit is not a universal full-scale benefit.

Property	Uniform PCM	Companded PCM
Input-domain step	Constant	Varies with amplitude
Small amplitudes	Coarser relative precision	Can receive finer precision
Main processing	Quantize and encode	Compress, quantize, encode, expand
Noise model	$\Delta^2/12$ for fixed step	Depends on local step and input statistics

Example 3.18 / A 30 dB full-scale sine requirement

Use $f_s = 8000$ samples/s. For uniform quantization,

$$1.76 + 6.02\nu \geq 30 \quad \Longrightarrow \quad \nu \geq 4.69.$$

Choose five bits and 32 levels. Then $r = 40$ kbit/s and ideal $B_{\min} = 20$ kHz. For a continuous mu-law model at large μ and sufficiently many levels, an approximate full-scale/high-level relation is

$$\mathrm{SQNR} \approx \frac{3L^2}{[\ln(1+\mu)]^2}.$$

For $\mu = 255$, its decibel form is approximately $20\log_{10} L - 10.11$. Requiring 30 dB gives $L \gtrsim 101.3$. Binary encoding therefore uses seven bits and 128 levels, with $r = 56$ kbit/s and $B_{\min} = 28$ kHz.

This approximation is not an exact G.711 performance formula. It shows why companding should not be described as always improving a full-scale sinusoidal SQNR. Its main purpose is useful performance over a wide input-amplitude range.

PULSE CODE MODULATION / 3.22

PCM storage and sampling margin

Count channels, extra bits and time explicitly.

Example 3.19 / Stereo recording with specified overhead

Two channels are sampled at 44.1 ksample/s with 16 bits per sample.

$$r_{\text{payload}} = 2(44100)(16) = 1411200\,\text{bit/s}.$$

For the exercise, assume 100 percent overhead: one extra bit for every payload bit. The total rate is $2(1411200) = 2822400$ bit/s. For one hour, multiply by 3600 seconds:

$$N = 2822400(3600) = 10160640000\,\text{bits}.$$

This is 10.16064 Gbit using decimal units. The assumed overhead is an exercise condition, not a complete physical-disc format model. A full-scale sine with 16-bit uniform quantization has predicted SQNR about 98.09 dB.

Example 3.20 / Television with a 20-percent sampling margin

Use $f_m = 4.5$ MHz and $L = 1024$. The word length is $\nu = 10$.

$$f_s = 1.2(2f_m) = 10.8\,\text{Msamples/s}.$$

$$r = 10(10.8) = 108\,\text{Mbit/s}.$$

Example 3.21 / A 15 kHz audio message

With 65536 levels, $\nu = 16$. Sampling 20 percent above Nyquist gives $f_s = 1.2(30000) = 36000$ samples/s. Therefore $r = 576$ kbit/s and ideal $B_{\text{min}} = 288$ kHz per channel, without overhead.

PULSE CODE MODULATION / 3.23

Questions for practice

Define, list, compare, explain and calculate.

Definitions and lists

1. Define PCM, quantization, quantization error and companding.
2. List the three basic PCM conversion operations in order.
3. List the receiver blocks and state the function of each.
4. Distinguish uniform and non-uniform quantization.

Explain why

Explain why: Why cannot the bit count be rounded up in a maximum-rate constraint?

Rounding up may make νf_s exceed the available channel rate. Use the largest feasible integer and then check the other requirements.

Explain why: Why must the input range be specified in a quantization problem?

The range determines the step size for a given number of levels. Without it, neither the rounding error nor the noise estimate is determined.

Numerical practice with answers

A. A 4-bit midrise quantizer covers 0 to 8 V. Quantize 3.2 V. **Answer:** $L = 16$, $\Delta = 0.5$ V, $i = 6$, $m_q = 3.25$ V, code 0110 and $e = -0.05$ V.

B. A PCM stream uses 6 bits at 10 ksample/s. Find bit rate, bit duration and ideal minimum bandwidth. **Answer:** 60 kbit/s, 16.667 μs and 30 kHz.

C. A full-scale sine requires SQNR of at least 50 dB. Find the minimum binary word length. **Answer:** $\nu \geq (50 - 1.76)/6.02 = 8.013$. Choose nine bits.

D. Draw a companding PCM system and explain why it can improve weak-signal resolution. Use the compressor and input-interval diagrams to support the answer.